

Medical Statistics

Lab 10: Integrated exercises with the Cleveland heart disease dataset

Introduction

In this final lab, you will work more independently with the Cleveland heart disease dataset. The analyses are specified, but you will decide how to carry them out, check the relevant output, and interpret the results clearly.

The exercises combine methods from the previous lab sessions:

- ANCOVA and regression diagnostics;
- logistic regression and odds ratios;
- causal diagrams and confounder adjustment;
- prediction model building.

Dataset description

The Cleveland heart disease dataset includes demographic, clinical, and diagnostic variables from patients evaluated for heart disease.

Key variables used in this lab:

- **age**: age in years
- **sex**: sex (Female, Male)
- **cp**: chest pain type (Typical angina, Atypical angina, Non-anginal pain, Asymptomatic)
- **trestbps**: resting blood pressure in mm Hg
- **chol**: serum cholesterol in mg/dl
- **fbs**: fasting blood sugar > 120 mg/dl (False, True)
- **thalach**: maximum heart rate achieved
- **target**: diagnosis of heart disease (Heart Disease, No Heart Disease)

Use the full dataset `heart_disease_cleveland.sav`.

Part A: ANCOVA

In this part, you will investigate whether resting blood pressure differs between males and females after accounting for age.

A1. Research question

Is there a difference in resting blood pressure (`trestbps`) between males and females after adjusting for age?

A2. Exploratory analysis

Before fitting a model:

1. Describe resting blood pressure by sex.
2. Make a scatterplot of resting blood pressure against age, using color or symbols to distinguish males and females.
3. Briefly comment on whether the relationship between age and resting blood pressure appears approximately linear.

A3. Additive ANCOVA model

Fit an additive ANCOVA model with `trestbps` as the outcome and `age` and `sex` as predictors.

Answer the following:

- the adjusted difference in resting blood pressure between males and females;
- the 95% confidence interval;
- the p-value for `sex`;
- a short interpretation in words.

A4. Model assumptions and interaction

Assess whether the additive ANCOVA model is reasonable.

1. Use a Q-Q plot to assess normality of the residuals.
2. Use a residuals-versus-fitted plot to assess homoscedasticity and linearity.
3. Fit an interaction model that allows the association between age and resting blood pressure to differ by sex.
4. Test the age-by-sex interaction term.
5. State whether the additive model seems appropriate.

Part B: Logistic regression and causal adjustment

In this part, you will investigate whether cholesterol is associated with the presence of heart disease.

B1. Preparing the outcome and exposure

For logistic regression, use heart disease as the event of interest. Code or define the outcome so that:

- 1/event = Heart Disease;
- 0/no event = No Heart Disease.

R reminder

If you are using R, create a binary outcome before fitting the logistic regression model:

```
heart_data <- heart_data |>
  mutate(target01 = ifelse(target == "Heart Disease", 1, 0))
```

B2. Unadjusted model

Fit a logistic regression model with heart disease status as the outcome and cholesterol as the only predictor.

Answer the following:

- the odds ratio for a 1 mg/dl increase in cholesterol;
- the corresponding odds ratio for a 10 mg/dl increase in cholesterol;
- the 95% confidence interval;
- the p-value;
- a short interpretation.

B3. Causal diagram

Consider the following hypothetical causal diagram for the relationship between cholesterol and heart disease.

The assumed arrows are:

- age -> cholesterol
- age -> heart disease

- sex -> cholesterol
- sex -> heart disease
- cholesterol -> heart disease
- heart disease -> chest pain type

Use this DAG to decide which variables should be included in the adjusted model.

Which variables should be included in the adjusted logistic regression model? Explain your choice using the DAG.

B4. Adjusted model

Fit the adjusted logistic regression model using the confounders identified from the DAG.

Answer the following:

- the adjusted odds ratio for a 1 mg/dl increase in cholesterol;
- the corresponding adjusted odds ratio for a 10 mg/dl increase in cholesterol;
- the 95% confidence interval;
- the p-value;
- whether the estimate changed compared with the unadjusted model;
- what this suggests about confounding.

Part C: Building a prediction model

In this part, you will build a prediction model for maximum heart rate achieved (**thalach**).

Candidate predictors:

- age
- sex
- cp
- chol
- fbs
- trestbps

C1. Candidate model

Start with a linear regression model with **thalach** as the outcome and the candidate predictors listed above.

For categorical predictors with more than two categories, evaluate the predictor as a whole rather than focusing on only one dummy coefficient.

C2. Model selection

Choose a variable selection strategy and apply it to build a final prediction model.

Answer the following:

- Which selection strategy did you use?
- Why is this strategy reasonable for a prediction model?
- Which predictors were retained in the final model?

C3. Prediction for a patient

Use the final model to predict maximum heart rate for the following patient:

- age: 62 years
- sex: female
- chest pain type: typical angina
- cholesterol: 240 mg/dl
- fasting blood sugar > 120 mg/dl: false
- resting blood pressure: 135 mm Hg

Only use predictors that are included in your final model.

Downloads

[Cleveland heart disease dataset \(SPSS format\)](#)